

Downscaled Climate Change Projections for South Africa (NEX-GDDP-CMIP6)

Executive summary

This report presents an analysis of projected climate extremes for South Africa using daily data from the NASA NEX-GDDP-CMIP6 dataset (Thrasher et al., 2022). The study was conducted using the open-source `nex-gddp-sa-tools` software package developed by SAEON (<https://github.com/SAEON/nex-gddp-sa-tools>). The workflow automates the downloading, processing, and analysis of climate model data across multiple emission scenarios (SSP2-4.5, SSP3-7.0, SSP5-8.5) and time horizons: near-term (2021–2040), mid-century (2041–2060), and late-century (2081–2100), relative to a 1995–2014 baseline.

Daily temperature and precipitation data were processed to calculate internationally recognised climate-extreme indices in accordance with World Meteorological Organisation (WMO) guidelines. While the analysis was informed by the `xclim` library (Bourgault et al., 2023), several indices required custom implementations due to compatibility and functionality constraints within existing `xclim` modules. These custom scripts ensured handling of South African regional datasets, calendar variations, and percentile-based thresholds.

Each climate model's first ensemble member (`r1i1p1f1`) was used to maintain equal representation in a multi-model ensemble. Ensemble averaging was performed at the end of the workflow to characterise the inter-model spread and reduce structural uncertainty.

Results show a marked intensification of temperature extremes and a reduction in cold events across all future periods. Warm-related indices, such as the Warm Spell Duration Index (WSDI) and the Percentage of Hot Days (TX90p), exhibit strong positive anomalies, especially in the Northern Cape, Limpopo, North West, and Free State provinces. In contrast, cold-related indices, including the Cold Spell Duration Index (CSDI) and Frost Days (FD), show widespread declines, indicating fewer and shorter cold events.

Precipitation-based indices highlight an increasing climate variability: eastern and coastal areas show a rise in extreme rainfall events, while central and western regions tend to be drier. By overlaying model outputs with South Africa's climate regions and district municipalities, the analysis provides spatially detailed insights into future climate risks. The results reinforce the need for regional adaptation planning, particularly in areas projected to experience sustained warming, reduced frost frequency, or more intense heatwaves.

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List of Acronyms, Units, and Technical Terms

Datasets, Frameworks, and Scenarios

Acronym / Term	Full Term / Description
AR6	Sixth Assessment Report: <i>the sixth major climate assessment by the Intergovernmental Panel on Climate Change (IPCC), published 2021–2023.</i>
CMIP6	Coupled Model Intercomparison Project Phase 6: <i>an international climate modelling collaboration that coordinates GCM simulations used for IPCC assessments.</i>
ETCCDI	Expert Team on Climate Change Detection and Indices: <i>WMO-led expert group defining standardised indices for monitoring climate and weather extremes.</i>
GCM	General Circulation Model: <i>a numerical model simulating atmospheric, oceanic, and land surface processes of the Earth's climate system.</i>
GMFD	Global Meteorological Forcing Dataset: <i>reference dataset used for bias correction and validation of global and regional climate model outputs.</i>
NEX-GDDP-CMIP6	NASA Earth Exchange Global Daily Downscaled Projections - CMIP6: <i>bias-corrected, statistically downscaled daily climate projections derived from CMIP6 models.</i>
ScenarioMIP	Scenario Model Intercomparison Project: <i>CMIP6 experiment providing projections for the SSP scenarios.</i>
SSP	Shared Socioeconomic Pathway: <i>scenario framework representing alternative future socioeconomic and greenhouse-gas trajectories (e.g. SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5).</i>

Tools, Software, and Methods

Acronym / Term	Full Term / Description
NetCDF	Network Common Data Form: <i>standard file format for storing array-oriented scientific data, widely used for climate and atmospheric datasets.</i>
nex-gddp-sa-tools	NEX-GDDP South Africa Tools: <i>an open-source Python package developed by SAEON for downloading, processing, and analysing CMIP6 downscaled data.</i>
xclim	Python Climate Analysis Library: <i>open-source library for computing WMO-defined climate indices (Bourgault et al., 2023, JOSS, 8 (85), 5415).</i>
YAML	YAML Ain't Markup Language: <i>a human-readable data format used for configuration files (.yaml).</i>

Variables and Units

Acronym / Term	Full Term / Description
%	Per cent: <i>unit used for percentile-based indices (e.g. TX90p, TN10p).</i>
°C	Degrees Celsius: <i>temperature unit used for analysis and visualisation.</i>
K	Kelvin: <i>SI unit of thermodynamic temperature.</i>
kg m ⁻² s ⁻¹	Kilograms per square metre per second: <i>unit for precipitation rate in model outputs.</i>
mm	Millimetres: <i>unit for precipitation totals.</i>
mm day ⁻¹	Millimetres per day: <i>unit for precipitation intensity.</i>
PR (pr)	Precipitation: <i>total daily precipitation rate (kg m⁻² s⁻¹), equivalent to mm day⁻¹ when integrated over 24 hours.</i>
r95p / r99p	95th and 99th Percentile Precipitation Thresholds: <i>thresholds defining very wet and extremely wet days.</i>
TASMAX (TX)	Maximum Near-Surface Air Temperature: <i>daily maximum 2 m air temperature, reported in Kelvin (K) and converted to °C.</i>
TASMIN (TN)	Minimum Near-Surface Air Temperature: <i>daily minimum 2 m air temperature, reported in Kelvin (K) and converted to °C.</i>
W m ⁻²	Watts per square metre: <i>unit of radiative forcing representing energy flux.</i>

Extreme Temperature-Based Climate Indices

Acronym / Term	Full Term / Description
CSDI	Cold Spell Duration Index: <i>annual number of days contributing to events where ≥ six consecutive days experience TN < 10th percentile (cold period).</i>
FD	Frost Days: <i>number of days when the minimum temperature (TN) < 0 °C.</i>
TN10p	Percentage of Cold Nights: <i>fraction of days when TN < 10th percentile.</i>
TN90p	Percentage of Warm Nights: <i>fraction of days when TN > 90th percentile.</i>
TNlt2	Cold Nights (< 2 °C): <i>number of days when TN < 2 °C.</i>
TNn	Annual Minimum of Daily Minimum Temperature: <i>coldest night of the year.</i>
TNx	Annual Maximum of Daily Minimum Temperature: <i>hottest night of the year.</i>
TX10p	Percentage of Cool Days: <i>fraction of days when TX < 10th percentile.</i>
TX90p	Percentage of Hot Days: <i>fraction of days when TX > 90th percentile.</i>
TXdTNd	Hot Days and Nights Combined: <i>number of events where both TX and TN exceed their respective 95th percentiles for d consecutive days.</i>
TXge30	Hot Days (≥ 30 °C): <i>number of days when TX ≥ 30 °C.</i>
TXgt50p	Days Above Median Temperature: <i>percentage of days when TX exceeds the 50th percentile.</i>

TXn	Annual Minimum of Daily Maximum Temperature: coldest day of the year.
TXx	Annual Maximum of Daily Maximum Temperature: hottest day of the year.
WSDI	Warm Spell Duration Index: annual number of days in periods of ≥ 6 consecutive days when TX > 90th percentile (warm period).

Extreme Precipitation-Based Climate Indices

Acronym / Term	Full Term / Description
CDD	Consecutive Dry Days: <i>maximum number of successive days with PR < 1.0 mm.</i>
CWD	Consecutive Wet Days: <i>maximum number of successive days with PR ≥ 1.0 mm.</i>
PRCPTOT	Total Precipitation from Wet Days: <i>total rainfall from days with PR ≥ 1 mm; used in ratio-based indices such as R95pTOT and R99pTOT.</i>
R10mm	Heavy Rain Days: <i>number of days with PR ≥ 10 mm.</i>
R20mm	Very Heavy Rain Days: <i>number of days with PR ≥ 20 mm.</i>
R95p	Precipitation from Very Wet Days: <i>total rainfall from days exceeding the 95th percentile.</i>
R95pTOT	Contribution from Very Wet Days: $100 \times R95p / PRCPTOT$; <i>proportion of rainfall from very wet days (%).</i>
R99p	Precipitation from Extremely Wet Days: <i>total rainfall from days exceeding the 99th percentile.</i>
R99pTOT	Contribution from Extremely Wet Days: $100 \times R99p / PRCPTOT$; <i>proportion of rainfall from extremely wet days (%).</i>
Rx1day	Maximum 1-Day Precipitation: <i>highest 1-day rainfall total (mm).</i>
Rx5day	Maximum 5-Day Precipitation: <i>highest 5-day rainfall total (mm).</i>
SDII	Simple Daily Intensity Index: <i>mean precipitation per wet day (mm day⁻¹).</i>
SPI	Standardised Precipitation Index: <i>a statistical measure of drought severity and duration (unitless).</i>

CMIP6 Global Climate Models (GCMs)

Acronym / Term	Full Term / Description
ACCESS-CM2	Australian Community Climate and Earth System Simulator: <i>Climate Model version 2</i>
ACCESS-ESM1-5	Australian Community Climate and Earth System Simulator: <i>Earth System Model version 1.5</i>
BCC-CSM2-MR	Beijing Climate Centre Climate System Model version 2, Medium Resolution
CanESM5	Canadian Earth System Model version 5
CMCC-CM2-SR5	Centro Euro-Mediterraneo sui Cambiamenti Climatici Climate Model 2, SR5
CMCC-ESM2	Centro Euro-Mediterraneo sui Cambiamenti Climatici Earth System Model 2
EC-Earth3	EC-Earth Consortium Model version 3
EC-Earth3-Veg-LR	EC-Earth version 3 with interactive vegetation, Low Resolution
FGOALS-g3	Flexible Global Ocean–Atmosphere–Land System Model grid-point version 3

GFDL-CM4	Geophysical Fluid Dynamics Laboratory Climate Model version 4
GFDL-CM4_gr2	Geophysical Fluid Dynamics Laboratory Climate Model version 4 (alternate grid gr2)
GFDL-ESM4	Geophysical Fluid Dynamics Laboratory Earth System Model version 4
IITM-ESM	Indian Institute of Tropical Meteorology Earth System Model
INM-CM4-8	Institute of Numerical Mathematics Climate Model version 4.8
INM-CM5-0	Institute of Numerical Mathematics Climate Model version 5.0
IPSL-CM6A-LR	Institut Pierre-Simon Laplace Climate Model version 6, Low Resolution
KACE-1-0-G	Korean Advanced Community Earth System Model version 1.0 (Global)
KIOST-ESM	Korea Institute of Ocean Science & Technology Earth System Model
MIROC6	Model for Interdisciplinary Research on Climate version 6
MPI-ESM1-2-HR	Max Planck Institute Earth System Model version 1.2, High Resolution
MPI-ESM1-2-LR	Max Planck Institute Earth System Model version 1.2, Low Resolution
MRI-ESM2-0	Meteorological Research Institute Earth System Model version 2.0
NESM3	Nanjing University of Information Science and Technology Earth System Model version 3
NorESM2-LM	Norwegian Earth System Model version 2, Low/Medium Resolution
TaiESM1	Taiwan Earth System Model version 1

Institutions and Organisations

Acronym / Term	Full Term / Description
DWS	Department of Water and Sanitation: <i>South African government department overseeing water resource management.</i>
IPCC	Intergovernmental Panel on Climate Change: <i>United Nations body assessing the science of climate change.</i>
MDB	Municipal Demarcation Board: <i>a statutory body determining and reviewing municipal boundaries in South Africa.</i>
NASA	National Aeronautics and Space Administration: <i>United States agency providing NEX-GDDP-CMIP6 data.</i>
SAEON	South African Environmental Observation Network: <i>developer of nex-gddp-sa-tools and lead organisation for this analysis.</i>
SAWS	South African Weather Service: <i>national meteorological authority providing climate data and regional classifications.</i>
WMO	World Meteorological Organisation: <i>UN specialised agency setting standards for climate observations and indices.</i>

Sectoral Classifications

Acronym / Term	Full Term / Description
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AFS	Agriculture and Food Security – indices relevant to agricultural productivity.
H	Hydrology – indices relevant to hydrological processes.
WRH	Water Resources and Health – indices associated with water availability and public-health impacts.

1. Introduction

This report presents an analysis of downscaled climate change projections for South Africa, derived from NASA’s Earth Exchange Global Daily Downscaled Projections (NEX-GDDP-CMIP6) dataset (Thrasher et al., 2022), based on simulations from the CMIP6 General Circulation Models (GCMs). The analysis focuses on extreme rainfall and temperature indices to provide an overview of trends in climate extremes across South Africa. The World Meteorological Organisation (WMO) has established a set of standardised climate indices, known as the Expert Team on Climate Change Detection and Indices (ETCCDI) indicators, used to monitor and compare changes in surface temperature and precipitation extremes.

The NEX-GDDP-CMIP6 dataset provides bias-corrected and statistically downscaled global climate projections derived from an ensemble of 35 CMIP6 GCMs (Eyring et al., 2016). The data are available at a spatial resolution of $0.25^{\circ} \times 0.25^{\circ}$ (approximately 25 km) and at a daily temporal resolution for the period 1950–2100, making them suitable for regional and national climate assessments. The projections cover four Shared Socioeconomic Pathways (SSPs), representing distinct greenhouse gas emission scenarios (O’Neill et al., 2016), and were produced in support of the IPCC’s sixth assessment report (AR6) (Calvin et al., 2023). The downscaled ScenarioMIPs are available through NASA’s Earth Exchange data portal ([link](#)).

Downscaling of GCM outputs addresses two main practical limitations. First, the coarse spatial resolution of global models, often around 100 km, restricts the ability to capture regional-scale climate processes and local variability (Gebrechorkos et al., 2023; Tapiador, 2020; Thrasher et al., 2022). Second, model outputs may exhibit statistical bias when compared to local observations (Gebrechorkos et al., 2023). The NEX-GDDP-CMIP6 products apply a bias-correction and spatial disaggregation process, in which GCM outputs are adjusted using the Global Meteorological Forcing Datasets (GMFD) and then interpolated to a 0.25° grid. This process preserves both temporal and inter-model variability (Gebrechorkos et al., 2023).

For this analysis, daily precipitation (pr), maximum near-surface air temperature (tasmax), and minimum near-surface air temperature (tasmin) were extracted from each model and experiment using a single ensemble member (r1i1p1f1) (Donat, 2023; Zhong, 2023). By selecting only the first ensemble member (r1i1p1f1) from each model, we ensure that each model contributes equally to a multi-model ensemble, avoiding over-representation of models with multiple realisations (Gooya et al., 2023).

Indices were computed using WMO-defined algorithms implemented in Python (based on the xclim software suite (Alexander et al., 2006)) for three 20-year time slices: near-term (2021–2040), mid-century (2041–2060), and late-century (2081–2100); relative to a 1995–2014 historical baseline. Multi-model ensemble means were calculated to reduce inter-model variability.

2. Methodology

Climate models are built to simulate the physical, chemical, and biological processes that govern the Earth's climate system (Stocker, 2011; Washington & Parkinson, 2005). These models discretise the climate system onto a three-dimensional grid, using numerical methods to simulate how the atmosphere, oceans, and land surface evolve from defined initial conditions (Tehrani et al., 2022). To represent the influence of external drivers, models incorporate both historical and future estimates of radiative forcings, such as changes in solar radiation and greenhouse gas levels, to simulate climate states over time (Tehrani et al., 2022). These forcings are prescribed following standard CMIP6 experiment protocols to ensure comparability across models and scenarios (Eyring et al., 2016)

2.1 Data Properties

The NASA Earth Exchange Global Daily Downscaled Projections (NEX-GDDP-CMIP6) provide daily, bias-corrected, and statistically downscaled climate projections derived from multiple CMIP6 global climate models (GCMs) (Thrasher et al., 2022). The data spans both the historical data (1950–2014) and future projections (2015–2100), aligned with the four principal Shared Socioeconomic Pathways (SSPs: SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5). Each SSP represents a distinct socio-economic trajectory and associated greenhouse gas concentration pathway, ranging from low-emission scenarios (SSP 1-2.6) to high-emission, fossil-fuel-intensive futures (SSP 5-8.5) (O'Neill et al., 2016).

The data are provided in Network Common Data (NetCDF) format at a spatial resolution of 0.25° (~25 km) and daily temporal frequency. Three primary climate variables were extracted for this study to compute a suite of 28 climate indices based on the WMO-defined ETCCDI framework (Zhang et al., 2011):

- Precipitation (pr): total daily precipitation rate ($\text{kg m}^{-2} \text{s}^{-1}$), equivalent to millimetres per day (mm day^{-1}) when integrated over 24 hours.
- Maximum near-surface air temperature (tasmax): daily maximum 2m air temperature (Kelvin), converted to degrees Celsius ($^{\circ}\text{C}$) for analysis, and
- Minimum near-surface air temperature (tasmin): daily minimum 2m air temperature (Kelvin), also converted to $^{\circ}\text{C}$.

The temporal analysis covers a historical baseline period (1995–2014) and three future time slices: near-term (2021–2040), mid-term (2041–2060), and far-term (2081–2100). The baseline was selected to align with the IPCC's AR6 assessment.

The spatial domain was limited to mainland South Africa, bounded by latitudes -35.0° to -21.0° and longitudes 16.0° to 33.0° . Model selection was based on data availability within the NEX-GDDP-CMIP6 archive, resulting in 25 GCMs. A single ensemble member (r1i1p1f1) was selected from each model to maintain equal weighting and isolate inter-model variability. The analysis also accounted for model-specific grid configurations (e.g., gr, gr1, gr2, gn), which identify the grid configurations used by each modelling centre. For example, *gr* refers to a regular-longitude grid, while *gn* denotes the model's native grid, which may be regular, irregular, or even curvilinear. Variables that conformed to a regular grid were selected to ensure consistent spatial alignment.

Table 1. List of the 25 CMIP6 Global Climate Models (GCMs) from the NEX-GDDP-CMIP6 dataset used in this report. The table provides each model's abbreviation and full name. The data is available from ([link](#))

No	Abbreviation	Model Name
1.	ACCESS-CM2	Australian Community Climate and Earth System Simulator: Climate Model version 2
2.	ACCESS-ESM1-5	Australian Community Climate and Earth System Simulator: Earth System Model version 1.5
3.	BCC-CSM2-MR	Beijing Climate Centre Climate System Model version 2, Medium Resolution
4.	CMCC-CM2-SR5	Centro Euro-Mediterraneo sui Cambiamenti Climatici Climate Model 2, SR5
5.	CMCC-ESM2	Centro Euro-Mediterraneo sui Cambiamenti Climatici Earth System Model 2
6.	CanESM5	Canadian Earth System Model version 5
7.	EC-Earth3	EC-Earth Consortium Model version 3
8.	EC-Earth3-Veg-LR	EC-Earth version 3 with interactive vegetation, Low Resolution
9.	FGOALS-g3	Flexible Global Ocean–Atmosphere–Land System Model grid-point version 3
10.	GFDL-CM4	Geophysical Fluid Dynamics Laboratory Climate Model version 4
11.	GFDL-CM4_gr2	Geophysical Fluid Dynamics Laboratory Climate Model version 4, alternate grid (gr2)
12.	GFDL-ESM4	Geophysical Fluid Dynamics Laboratory Earth System Model version 4
13.	IITM-ESM	Indian Institute of Tropical Meteorology Earth System Model
14.	INM-CM4-8	Institute of Numerical Mathematics Climate Model version 4.8
15.	INM-CM5-0	Institute of Numerical Mathematics Climate Model version 5.0
16.	IPSL-CM6A-LR	Institut Pierre-Simon Laplace Climate Model version 6, low resolution
17.	KACE-1-0-G	Korean Advanced Community Earth System Model version 1.0, Global
18.	KIOST-ESM	Korea Institute of Ocean Science & Technology Earth System Model
19.	MIROC6	Model for Interdisciplinary Research on Climate version 6
20.	MPI-ESM1-2-HR	Max Planck Institute Earth System Model version 1.2, High Resolution
21.	MPI-ESM1-2-LR	Max Planck Institute Earth System Model version 1.2, Low Resolution
22.	MRI-ESM2-0	Meteorological Research Institute Earth System Model version 2.0

23.	NESM3	Nanjing University of Information Science and Technology Earth System Model version 3
24.	NorESM2-LM	Norwegian Earth System Model version 2, Low/Medium Resolution
25.	TaiESM1	Taiwan Earth System Model version 1

The Shared Socioeconomic Pathways (SSPs) (see table 2) outline different global development paths, each linked to specific levels of radiative forcing and projected temperature rises by the end of the 21st century (O'Neill et al., 2016). The following SSPs were analysed: SSP2-4.5, representing a moderate, middle-of-the-road scenario characterised by modest sustainability efforts and trends mostly aligned with historical patterns, resulting in approximately 4.5 W/m² forcing and an estimated warming of 2.5–3 °C. SSP3-7.0 depicts a global landscape characterised by regional competition, limited international cooperation, and higher population growth, resulting in approximately 7.0 W/m² forcing and a forecasted warming of 3.5–4 °C (Meinshausen et al., 2020). And, SSP5-8.5, which presents a development scenario heavily dependent on fossil fuels, rapid economic growth, energy-intensive lifestyles, and significant greenhouse gas emissions, resulting in roughly 8.5 W/m² forcing and warming of about 4.5–5 °C (Meinshausen et al., 2020).

Table 2. CMIP6 Shared Socioeconomic Pathways (SSPs), along with their associated forcing levels by 2100 (O'Neill et al., 2016).

SSP Name	Radiative Forcing (W/m ² by 2100)	Climate Target	Socioeconomic Narrative
SSP1-1.9	~1.9 W/m ²	1.5°C goal (Paris Agreement)	Sustainability-focused world: Rapid shift to green energy, low inequality, strong institutions.
SSP2-4.5	~4.5 W/m ²	~2.5–3 °	Middle of the road: Modest progress on sustainability; trends follow historical patterns
SSP3-7.0	~7.0 W/m ²	~3.5–4 °C	Regional rivalry: a fragmented world, weak international cooperation, and high population growth.
SSP5-8.5	~8.5 W/m ²	~4.5–5 °C	Fossil-fueled development: Rapid economic growth, energy-intensive lifestyles, high GHG emissions.

2.2 Study Area

The study area is South Africa, defined by a geographical bounding box of: latitude -35.0 to -21.0 and longitude 16.0 to 33.0. This report analysed surface temperature and precipitation indices across 23 climatic regions and 52 district municipalities within these boundaries.

The municipal district boundaries are defined and maintained by the Municipal Demarcation Board (MDB), who periodically review and updates them every five years, following local government elections (Municipal Demarcation Board, 2018). The 23 climatic regions were provided by the South African Weather Service (SAWS) (South African Weather Service, 2015) and delineate areas within South Africa that share similar climatic conditions and biophysical environments. Refer to Figures 1 and 2 below.

South African District Municipalities (2018)

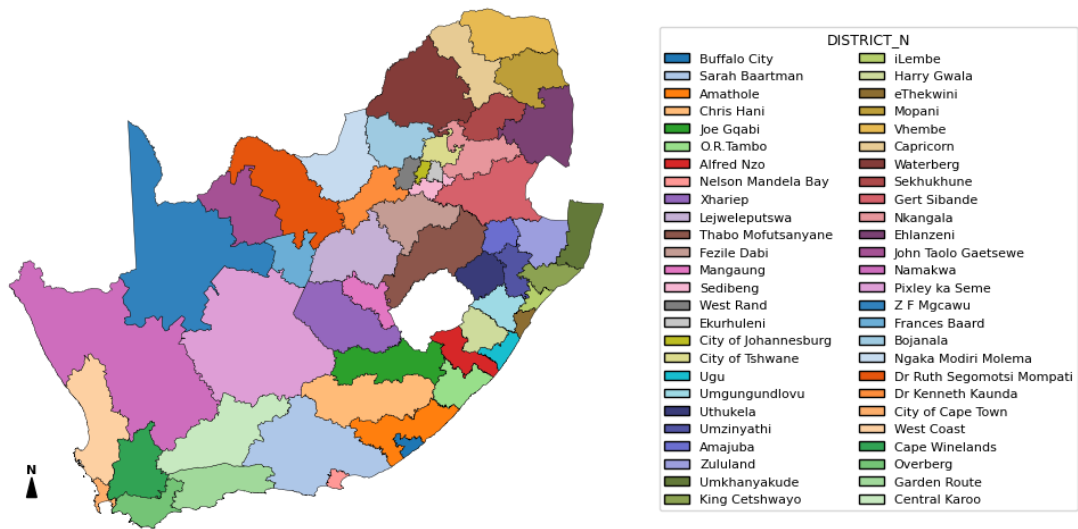
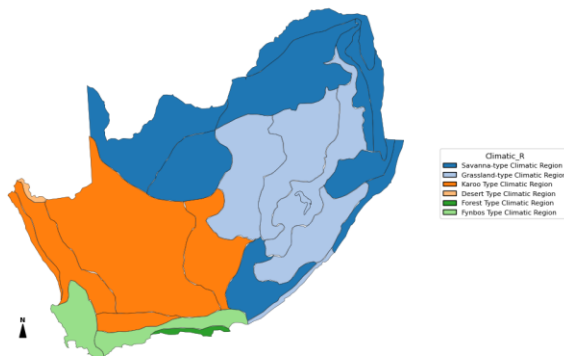


Figure 1. South African District Municipality Boundaries of South Africa (2018)

South African Climate Regions (Climate_R)



South African Climate Regions: Vegetation Biomes

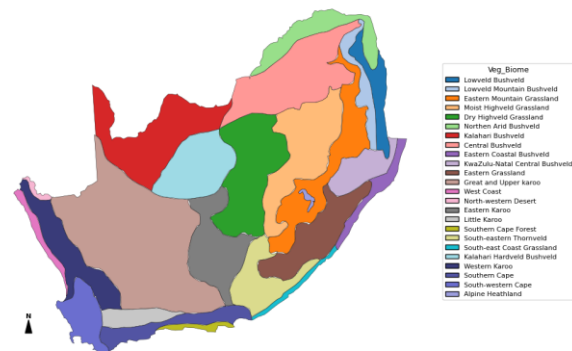


Figure 2. South African Climate Regions & Vegetation Biomes: These regions share the same temperature, rainfall, and ecosystem structure, forming the basis for climate and biodiversity classification across the country.

2.3 Data Processing

Data Processing and analysis were conducted using the open-source nex-gddp-sa-tools software package, developed by SAEON (freely available on GitHub ([LINK](#))). The workflow involves:

- (1) exploring available models, variables, and scenarios;
- (2) automatically downloading and verifying daily climate data;
- (3) checking the integrity of downloads;
- (4) calculating climate indices from precipitation and temperature data;
- (5) visualising anomalies relative to the historical baseline.

All scripts are configured in YAML files (download_config.yml, climate_indices_config.yml, and plot_config.yml) that specify models, variables, time periods, output directories, and plotting configurations. The workflow is transparent and version-controlled, ensuring reproducibility.

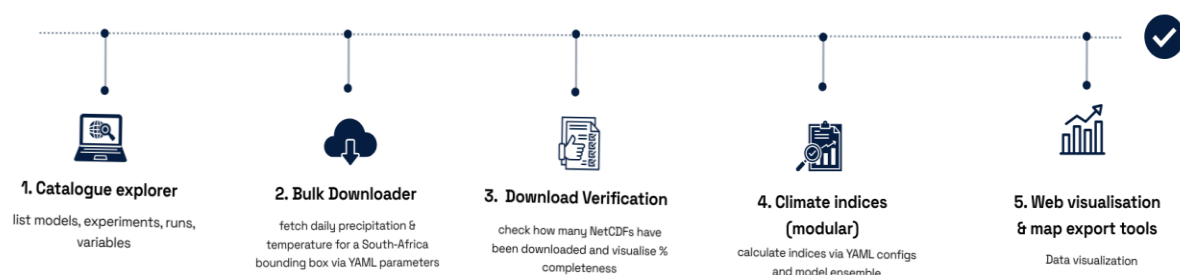


Figure 3. Operational workflow for data processing, analysis, and visualisation.

2.3.1 Data Acquisition and Calculation

Daily climate data were obtained directly from the NASA servers through the following scripts:

- **run_downloads.py:** This script automatically connects to NASA’s public THREDDS servers and downloads all required climate model files for South Africa. It keeps a record of which models, scenarios, and years were downloaded and organises the files by variable, model name, and experiment (for example, SSP2-4.5).
- **download_sa_subset.py:** Sometimes it is helpful to download only a small portion of the data, for example, a single model or scenario, for testing purposes.
- **verify_downloads.py:** This script checks that everything is correct. It compares what was expected (e.g., 365 daily files per year and model) with what was actually downloaded. If anything is missing, duplicated, or corrupted, it lists the problems.
- **run_climate_indices.py:** This script takes precipitation and temperature data and calculates the climate-extreme indicators. The calculations follow World Meteorological Organisation (WMO) standards, ensuring results are directly comparable with other national and international studies.
- **plot_time_slices_by_bioregion.py** and **plot_time_slices_by_municipality.py:** These scripts create the maps used in the report. They show how each climate index changes over time (near-term, mid-century, and late-century) and across scenarios. Colours and legends are standardised (and configurable).

Daily precipitation and temperature data were processed using scripts adapted from the xclim (Bourgault et al., 2023) climate indicators library. Tables 3 and 4 below summarise the set of indices evaluated in this report, outlining their definitions and interpretive context.

Table 3. WMO-ETCCDI: Extreme Temperature Indices Assessed

Short name	Long name	Definition	Plain language description	Units	Sector*
CSDI	Cold spell duration indicator	Annual number of days contributing to events where six or more consecutive days experience $TN < 10$ th percentile	Number of days contributing to a cold period (where the period has to be at least 6 days long)	days	H, AFS

FD	Frost Days	Number of days when $TN < 0^{\circ}C$	Days when the minimum temperature is below $0^{\circ}C$	days	H, AFS
TN10p	Amount of cold nights	Percentage of days when $TN < 10$ th percentile	Fraction of days with cold nighttime temperatures	%	All
TN90p	Amount of warm nights	Percentage of days when $TN > 90$ th percentile	Fraction of days with warm nighttime temperatures	%	All
TNlt2	TN below $2^{\circ}C$	Number of days when $TN \geq 2^{\circ}C$	Days when the minimum temperature is below $2^{\circ}C$	days	AFS
TNn	Min TN	Coldest daily TN	Coldest night	$^{\circ}C$	AFS
TNx	Max TN	Warmest daily TN	Hottest night	$^{\circ}C$	All
TX10p	Amount of cool days	Percentage of days when $TX < 10$ th percentile	Fraction of days with cool daytime temperatures	%	All
TX90p	Amount of hot days	Percentage of days when $TX > 90$ th percentile	Fraction of days with hot daytime temperatures	%	All
TXdTnd	User-defined consecutive number of hot days and nights	Annual count of d consecutive days where both TX & 95th percentile and TN & 95th percentile	events	days	H, AFS, WRH
TXge30	TX of at least $30^{\circ}C$	Number of days when $TX \geq 30^{\circ}C$	Days when maximum temperature is at least $30^{\circ}C$	days	H, AFS
TXgt50p	Fraction of days with temperatures above the median	Percentage of days where TX; 50th percentile	Fraction of days with above-median temperature	%	H, AFS, WRH
TXn	Min TX	Coldest daily TX	Coldest day	$^{\circ}C$	All
TXx	Max TX	Warmest daily TX	Hottest day	$^{\circ}C$	AFS
WSDI	Warm spell duration indicator	Annual number of days contributing to events where six or more consecutive days experience $TX > 90$ th percentile	Number of days contributing to a warm period (where the period has to be at least 6 days long)	days	H, AFS, WRH

* AFS (Agriculture and Food Security), H (hydrology), and WRH (Water Resources and Health)

Table 4. WMO-ETCCDI: Extreme Precipitation Indices Assessed

Short name	Long name	Definition	Plain language description	Units	Sector*
CDD	Consecutive Dry Days	Maximum number of consecutive dry days (when $PR < 1.0$ mm)	Longest dry spell	days	H, AFS, WRH
CWD	Consecutive Wet Days	Maximum annual number of consecutive wet days (when $PR \geq 1.0$ mm)	The longest wet spell	days	All
R10mm	Number of heavy rain days	Number of days when $PR \geq 10$ mm	Days when rainfall is at least 10mm	days	All
R20mm	Number of very heavy rain days	Number of days when $PR \geq 20$ mm	Days when rainfall is at least 20mm	days	AFS, WRH
SPI	Standardised Precipitation Index	Measure of "drought" using the Standardised Precipitation Index on time scales of 3, 6 and 12 months. See McKee et al. (1993) and the WMO SPI User guide (World Meteorological Organisation, 2012) for details.	A drought measure specified as a precipitation deficit	unitless	H, AFS, WRH
R95pTOT	Contribution from very wet days	$100 \times r95p / PRCPTOT$	Fraction of total wet-day rainfall that comes from very wet days	%	AFS, WRH

R99pTOT	Contribution from extremely wet days	$100 \times r_{99p} / \text{PRCPTOT}$	Fraction of total wet-day rainfall that comes from extremely wet days	%	AFS, WRH
R95p	Total annual PR from heavy rain days	Annual sum of daily PR & 95th percentile	Amount of rainfall from very wet days	mm	All
R99p	Total annual PR from very heavy rain days	Annual sum of daily PR & 99th percentile	Amount of rainfall from extremely wet days	mm	All
Rx1day	Max 1-day PR	Maximum 1-day PR total	The maximum amount of rain that falls in one day	mm	All
Rx5day	Max 5-day PR	Maximum 5-day PR total	The maximum amount of rain that falls in five consecutive days	mm	All
SDII	Daily PR intensity	Annual total PR divided by the number of wet days (when total PR ≥ 1.0 mm)	Average daily wet-day rainfall intensity	mm/day	All

* AFS (Agriculture and Food Security), H (hydrology), and WRH (Water Resources and Health)